

AADI KAUL

Software Engineer · Data, AI & Full-Stack

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Available: Aug 31, 2026 – Jan 22, 2027

Third-year Computer Science student at SUTD who ships production software, not just class projects: sole engineer on live fintech and government systems, and builder of ML-powered fraud detection, RAG-based LLM tools, and full-stack data platforms end to end, from REST APIs and ETL pipelines to the dashboards that make the data usable. Looking to bring that ownership to a data engineering, data analytics, AI/ML engineering, or software engineering internship.

TECHNICAL SKILLS

Languages: Python · Java · SQL · JavaScript · TypeScript · C++
Data, ML & Analytics: Pandas · NumPy · scikit-learn · TF-IDF · RAG · Isolation Forest · SQL Analytics · Tableau · Power BI
Backend & Frontend: FastAPI · Flask · Node.js · REST APIs · async SQLAlchemy · React.js · Next.js · Tailwind CSS
Cloud, DevOps & AI/LLM: AWS · Docker · Docker Compose · CI/CD · Linux · Git · LLM APIs · Prompt Engineering · Vector Similarity Search

EXPERIENCE

Software Developer May 2026 – Present
PayPal Career Academy · Singapore

- Selected from a competitive applicant pool for PayPal's Career Academy, a software engineering programme embedded within its global fintech engineering organisation.
- Building full-stack features under structured senior engineer mentorship, working within **Agile/Scrum** sprints and Git-based code review to gain hands-on exposure to enterprise-scale system design and production deployment workflows in a regulated fintech environment.

Software & Business Automation Intern Sept 2025 – Jan 2026
Central Provident Fund Board (CPF) · Singapore

- Sole engineer** on 5 production full-stack web apps (React, Flask, SQLite) automating end-to-end government workflows, eliminating manual processes and saving an estimated 10+ staff hours per week across teams.
- Built **Python ETL pipelines** integrating enterprise backend systems via REST APIs, improving data accuracy and reducing operational overhead.
- Owned the **full delivery lifecycle**, from stakeholder interviews and process mapping through to production rollout, in a regulated government environment.
- Engineered **RPA-driven automation workflows** with comprehensive test suites.

IT Trainee Mar 2024 – Jun 2024
Altrocks Tech Pte Ltd · Singapore

- Developed a **vendor-liaison tracking system** for client Resorts World Sentosa, digitising vendor-operations coordination for their casino; also resolved hardware, software, and network issues across client environments, authoring runbooks that cut onboarding time.

EDUCATION

Singapore University of Technology and Design 2024 – 2028
BEng Computer Science · Year 3 · Expected 2028

- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Computer Security, Computational Design.
- Participated in hackathons building full-stack, software engineering, and machine learning projects under time constraints, alongside mixed-methods data analysis (Python, Pandas) for elderly UX research presented to a faculty panel.

SELECTED PROJECTS

FraudIQ · ML-based fraud detection system
Flags anomalous transactions using **Isolation Forest** anomaly detection over engineered features, achieving **0.97 ROC-AUC** and 74.9% precision on flagged transactions (3 in 4 flags are real fraud), served via a FastAPI backend with a React dashboard

HDB Resale Intelligence Platform · Full-stack data engineering platform
FastAPI + async SQLAlchemy ETL pipeline processing **4,000+ real HDB resale transactions across 26 towns** (avg. resale price \$664K), feeding a 4-tab interactive analytics dashboard (Overview, Explorer, Price Map, Pipeline).

EarningsIQ · RAG-based financial report Q&A assistant
Chunking + **TF-IDF retrieval** pipeline over company annual reports (PDF upload supported); retrieves top-4 passages in **under 1ms** via pure in-memory search, no vector-DB network calls, powered by Llama 3.1 via Groq for grounded question-answering.

PriceWatch SG · Real-time cost-of-living tracker
Aggregates and visualises pricing data across **15 everyday items** drawn from SingStat's monthly survey of 150+ outlets, surfacing Singapore cost-of-living trends over time.

AWARDS

Best Sustainable Design (SUTD) · EAGLES Leadership Award (MOE) · Edusave Academic Award (MOE) · Service to School Award (MOE)